Case F

Mr. Walters — Chronic Non-Specific Low Back Pain | Working Dx: CNSLBP with Central Sensitisation & Fear-Avoidance Behaviour

Musculoskeletal Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Generalised lumbar stiffness across all planes
- Poor transversus abdominis and multifidus activation
- Reduced cardiovascular fitness and physical deconditioning
- Central sensitisation — amplified, disproportionate pain response
- Reduced social participation and daily activity
- Loss of retirement role and identity

■■ RED FLAGS

- "You mentioned you believe your spine is 'crumbling' — can you tell me more about what you think is happening inside your back?"
- "You've stopped walking your dog — can you describe what you're worried will happen if you do walk, or lift, or exercise?"
- "How is your sleep — do you wake because of pain, and how is your mood and energy generally?"
- Cauda Equina Syndrome (Bowel / Bladder Dysfunction, Saddle Anaesthesia)
- Action: Immediate emergency department referral. Do not proceed with assessment if confirmed.
- Unexplained Weight Loss / Night Pain / Systemic Symptoms (Malignancy or Infection)
- Action: If weight loss, night sweats, night pain, fever, or age-appropriate cancer screening is overdue — urgent GP referral with bloods (FBC, CRP, PSA, calcium, protein electrophoresis).
- History of Osteoporosis / Significant Trauma (Vertebral Fracture Risk)

■ PATHOPHYSIOLOGY (KEY POINTS)

- In this state, the dorsal horn neurones of the spinal cord become sensitised such that non-noxious stimuli (normal movement, light touch, routine posture changes) are amplified and perceived as painful.
- This explains why Mr.
- Walters experiences constant pain despite no structural findings on imaging.
- The result is a pain system that is "turned up" independently of tissue state.
- Fear-avoidance behaviour operates as a powerful amplifier of this central sensitisation.
- Walters' belief that his spine is crumbling activates the limbic system, increasing amygdala-mediated threat appraisal.

■ DIFFERENTIAL DIAGNOSES

1. Chronic Non-Specific LBP with Central Sensitisation & Fear-Avoidance

2. Lumbar Spondylosis / Facet Joint Osteoarthritis

3. Lumbar Canal Stenosis (Neurogenic Claudication)

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Observation Findings

Posture: Observe for loss of lumbar lordosis, increased thoracic kyphosis (from years of sustained driving posture), or lateral shift Movement quality: Observe for pain behaviours (guarding, grimacing, bracing), hesitation before movement, catastrophic movement patterns — often more revealing than the ROM itself Gait:

Test 1 — Tampa Scale of Kinesiophobia (TSK-17)

What it measures A validated 17-item self-report questionnaire measuring pain-related fear of movement and (re)injury. Scores range from 17 to 68 — higher scores indicate greater kinesiophobia. How to perform

1. Provide the patient with the Tampa Scale questionnaire — it can be completed in the waiting room or at the start of the session.

2. Patient rates each of 17 statements on a 4-point Likert scale (1 = strongly disagree, 4 = strongly agree). 3. Score the total and note subscale patterns: "harm" (fear of pain/injury) and "activity avoidance" subscales identify dominant themes. 4. Interpret in context: scores >37 indicate clinically significant kinesiophobia. Positive finding TSK score >37 (confirmed "high" in Mr. Walters' case)

Test 3 — Repeated Movement Assessment (McKenzie Method — Directional Preference)

What it measures Whether repeated movements in a specific direction (flexion or extension) centralise, peripheralise, or have no effect on symptoms — identifies if a mechanical or directional component exists within this chronic presentation. How to perform 1. Perform 10 repeated end-range lumbar extensions in standing (or prone press-ups if tolerated) while monitoring symptom response after each set. 2. Then perform 10 repeated lumbar flexions in seated or standing and monitor again.

■ TREATMENT PLAN

- "What has your GP or specialist told you about your back? Have you been told what's wrong, and have your scans been explained to you?"

• Action: Screen with FRAX tool; request DXA scan if indicated; modify manual therapy approach accordingly.
• Progressive Neurological Deficit (New Leg Weakness, Sensory Loss)
• Action: Urgent neurological review and MRI lumbar spine.
• Psychosocial / Yellow Flags
• Significant Yellow Flags — Primary Drivers of Chronicity
• High kinesiophobia (Tampa Scale):
• Mr. Walters fears that movement will cause further spinal damage — this is the most significant barrier to recovery and must be the primary target of intervention
• Catastrophising:
• Activity avoidance:
• Cessation of dog walking and all lifting has led to deconditioning, social withdrawal, and reinforcement of the sick role
• Retirement identity loss: